

The Q-MONSTAR consortium: advancing fault-tolerant quantum computing for precision oncology



We present the Q-MONSTAR (Quantum-MONSTAR) from SCRUM-Japan, an academic–industry consortium established to verify the potential of quantum computing and develop fault-tolerant quantum computing (FTQC) applications in cancer treatment. SCRUM-Japan is a nationwide large-scale, industry–academia cancer genome screening project, coordinated by the National Cancer Center Japan. To our knowledge, Q-MONSTAR represents one of the first global proof-of-concept platforms for FTQC embedded within an ongoing clinical research infrastructure that leverages large-scale clinically validated multi-omics data.

Advances in precision oncology have necessitated a molecular-level understanding of cancer biology¹. However, many central problems – including protein conformational dynamics, accurate prediction of quantum effects in drug binding, and the combinatorial complexity of genomic mutations – remain computationally intractable for conventional high-performance computing (HPC). For example, exact quantum-mechanical calculations of the binding free energy between cancer-related proteins and small-molecule compounds are estimated to require computational times that far exceed the practical limits of HPCs (over 1,000 years)². These computational barriers impede the realization of accurate precision oncology and effective drug discovery.

FTQC has the potential to address these barriers by leveraging fundamental principles of quantum mechanics to enable powerful calculations that assess an exponentially larger computational space beyond what is classically feasible³. By using quantum superposition and entanglement, calculations that require exponential time on HPCs can be solved in a far more scalable and efficient way⁴. Notably, the photonic architecture developed by the quantum computing company PsiQuantum combines manufacturability and scalability to the million physical qubit regime

and beyond, enabling large-scale error correction necessary to execute deep quantum circuits for FTQC⁵.

Q-MONSTAR is led by the National Cancer Center Japan, together with pharmaceutical and biotechnology companies, and PsiQuantum, headquartered in California, as the leading quantum computing enterprise for advancing FTQC technologies and their potential applications. It is developed on the SCRUM-MONSTAR large-scale database, which contains integrated clinical and multi-omics data from over 30,000 patients with cancer, including whole-genome sequencing, whole-exome sequencing, whole-transcriptome sequencing, single-cell analyses, digital pathology, radiomics and high-quality clinical data⁶. So far, the database has facilitated the approval of 22 diagnostic and 24 therapeutic agents, providing a reliable foundation for validating proof-of-concept quantum computing approaches.

We focus on three major domains in which FTQC may contribute to precision oncology:

First, we target predictive analysis of drug properties based on accurate energy estimation and molecular dynamics simulations. To this end, we use quantum phase estimation (QPE)-based algorithms to calculate cancer-relevant protein–drug interactions at the quantum level and develop integrated molecular dynamics approaches to leverage both QPE-based and time-evolution-based approaches to improve the current methodologies⁷. Crucially, selecting biologically relevant protein conformations during initial state preparation is essential, as this choice governs the validity of all downstream predictions from binding energetics to functional outcomes. Artificial intelligence (AI)-based generative models and multiscale approaches can identify appropriate conformational ensembles where structural data are incomplete.

Second, modeling of chemically complex therapeutic mechanisms has become a crucial frontier in precision oncology. The development of covalent and non-covalent inhibitors that target biologically intricate enzymatic

systems, such as epigenetic regulators, including DNA methyltransferases and TET dioxygenases, has emerged as a promising strategy for potent and selective therapeutic modulation⁸. Unlike reversible inhibitors, covalent agents induce long-lasting functional changes; however, accurately predicting their behavior, particularly within catalytically active environments, remains a major computational challenge.

These systems frequently exhibit strong electronic correlation, multireference character, and complex reaction pathway, that are inadequately captured by traditional single-reference quantum chemical approaches, often failing to capture essential physics. Although many-body electronic structure methods can, in principle, achieve chemically accurate predictions, their computational cost increases factorially with the system size. Moreover, biologically relevant systems often require active spaces that exceed the practical limits of classical computation.

To overcome these challenges, multiscale quantum embedding frameworks provide a promising strategy by combining the high-fidelity quantum mechanical treatment of chemically crucial regions with scalable approximations of the surrounding environment. Such approaches enable more resource-efficient simulations while retaining the essential chemical accuracy. However, these advanced methods are limited by classical computational scaling. FTQC has the potential to integrate these approaches into a generalizable computational framework, enabling quantum-accurate modeling of complex biomolecular systems that are currently inaccessible. By enabling the precise chemical evaluation of electronically intricate states, including those relevant to covalent inhibition, FTQC could substantially advance the rational design of next-generation targeted therapies without reliance on system-specific assumptions⁹.

Third, advances in quantum algorithms may enable new approaches to the analysis of clinical data. This workstream seeks to determine

whether specific classes of clinical problems exhibit sufficient structure to permit efficient data loading and the extraction of clinically meaningful outputs while preserving a quantum advantage. Whether such conditions can be satisfied for real-world clinical datasets remains an open research question that we aim to address¹⁰. From a clinical perspective, defining which clinical questions merit quantum treatment is itself a central scientific objective.

Q-MONSTAR also incorporates solutions to practical implementation-related challenges. These include protecting the confidentiality of data, complying with legal and ethical constraints, managing and standardizing data quality, building a framework to protect intellectual property in collaborative research, and laying the foundation for international collaboration through our comprehensive implementation framework. Specifically, we must establish a protocol to verify the clinical validity of FTQC results and optimize the environment to ensure that actual FTQC operations can begin before the end of this decade.

The Q-MONSTAR initiative has the potential to revolutionize precision oncology by enabling the accurate simulation of drug targets with complex electronic structure previously beyond classical computational reach. FTQC could facilitate the rational design of covalent and noncovalent inhibitors tuned to the quantum-level behavior of disease-relevant proteins, accelerating therapeutic development with enhanced potency and selectivity. By integrating FTQC-based simulations into the drug-development pipeline, Q-MONSTAR seeks to transition oncological drug discovery from empirical screening into a

mechanistically grounded design framework, while emphasizing the parallel need for technological progress and appropriate ethical, legal and societal safeguards. Q-MONSTAR endeavors to introduce a new methodology to global precision oncology innovation and create a brighter future for as many patients with cancer as possible.

Mitsuho Imai¹, Riu Yamashita², Takao Fujisawa^{1,3}, Hiroki Sokuoka¹, Izumi Miki¹, Yasutoshi Sakamoto¹, Toshihiro Misumi⁴, Borna Zandkarimi⁵, Philipp T. Ernst⁵, Hideaki Bando^{1,6} & Takayuki Yoshino^{1,7}✉

¹Translational Research Support Office, National Cancer Center Hospital East, Kashiwa, Japan. ²Division of Translational Informatics, Exploratory Oncology Research and Clinical Trial Center, National Cancer Center Hospital East, Kashiwa, Japan.

³Department of Head and Neck Medical Oncology, National Cancer Center Hospital East, Kashiwa, Japan. ⁴Department of Data Science, National Cancer Center Hospital East, Kashiwa, Japan. ⁵PsiQuantum, Palo Alto, CA, USA. ⁶Department of Gastroenterology and Gastrointestinal Oncology, National Cancer Center Hospital East, Kashiwa, Japan. ⁷Department of Global Oncology, National Cancer Center Hospital East, Kashiwa, Japan.

✉e-mail: tyoshino@east.ncc.go.jp

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Competing interests

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